

Consider the following optimization primal problem.

- Assess if it has an optimal solution, if it is unbounded, or if it is infeasible.
- Write the Lagrangian, the dual function and the dual problem.
- Assess if the dual problem has an optimal solution, if it is unbounded, or if it is infeasible.

$$\min_{x \in \mathbb{R}^2} 4x_1 + 2x_2$$

subject to

$$\begin{aligned}x_1 - x_2 &= 1, \\x_1 &\geq 0, \\x_2 &\geq 0.\end{aligned}$$